ABHIJIT MITRA

Founder & CTO, TorqueX.ai · AI Engineering Leader — Agent Orchestration, LLMs & RAG

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PROFESSIONAL SUMMARY

Engineering leader with **10+ years** building and scaling **AI-powered products** — now founder of an **agent-orchestration platform**, with prior depth leading an AI document-intelligence product and distributed systems across logistics, travel, and content.

- Founder & CTO of **TorqueX.ai**, building **Agent X — an "Agent OS"** of custom, production AI agents. Own **agent orchestration end-to-end** — LLMs + **tool/function calling** + persistent, shared **agent memory** + **human-in-the-loop guardrails** + **evals** — shipping autonomous agents that research, act on live signals, and escalate safely.
- At Parspec, led engineering for an **AI-powered product-data platform** — large-scale document parsing, OCR, and ML extraction pipelines with **confidence scoring and human-in-the-loop review** — driving the business from inception to **$7M+ ARR** and scaling the team from **0 to 40+ engineers**.
- Architected and shipped an **end-to-end agentic development pipeline** — multi-agent orchestration with human-in-the-loop quality gates across code generation, review, testing, and deployment — increasing delivery velocity by **80%+**.
- At Locus.sh, engineered logistics-optimization systems scaling to **300M–1B+ API requests/day** during peak events, serving enterprises like **Myntra, BigBasket, Delhivery, Bukalapak, and Unilever**.
- Earlier, built **content-delivery platforms** for global publishers (**McGraw Hill, Oxford University Press, Pearson**) serving **15M+ users across 32 countries**.
- Drove **76% cloud cost reduction** ($288K → $70K/month) and led **SOC 2 Type I & II** with enterprise customer security assessments — relevant to scaling AI infrastructure cost-efficiently for enterprise buyers.
- Proven across successful startup journeys — **Tekion** (seed → **$4B**), **Spotnana** (early → **$1.2B unicorn**), **Locus.sh** (**acquired by IKEA/Ingka Group**), and **Parspec** (inception → **$7M+ ARR**).

PROFESSIONAL EXPERIENCE

Founder & CTO · **TorqueX.ai** Jul 2026 – Present

Bengaluru, India · Building Agent X — an "Agent OS" that runs a business as custom AI agents you own

- Founded TorqueX to build **Agent X** — an "Agent OS" that runs a business end-to-end (site, marketing, cold outreach, meetings, support) as custom, production AI agents. Own the full stack: product, **agent orchestration**, infra, and go-to-market.
- Architected the **agent-orchestration layer** — LLMs + **tool/function calling** + persistent, shared **agent memory** + **human-in-the-loop guardrails** + **evals** — so agents plan multi-step work, act autonomously, and escalate to a human when confidence is low.
- Shipped the first live agent — an autonomous **Outreach / GTM agent** (app.torquex.ai) that researches accounts, runs multi-step email, LinkedIn & phone sequences with follow-ups, and books meetings — escalating from email to **autonomous voice**.
- Built **reply intelligence** that reads inbound mail and **takes action** — an LLM classifier (few-shot + **structured output**) plus deterministic rules that suppress opt-outs globally, handle bounces, extract successor contacts from "left the company" replies and auto-enroll them, and escalate genuine buying signals to a human; every auto-action is **logged and reversible**.

- Built **Kova**, a vertical AI agent for construction teams that turns an incoming request into a priced quote (RFQ → BOM → quote → invoice) — live with real users.
- Runs on an async **FastAPI** backend, Postgres, Redis-backed workers (send, warmup, reply-polling, health monitoring), and **Cloudflare Workers**; models served via Claude and OpenRouter.

Director of Engineering · Parspec Jan 2025 – May 2026

Bengaluru, India · AI-powered product data platform for the building industry

- Led the end-to-end engineering journey, scaling the company from inception to **$7M+ ARR** through strong product and technology execution.
- Built and scaled the engineering organization from **0 to 40+ engineers**, including **4 Engineering Managers and 1 Application Security Lead**, establishing robust systems across development, security, and QA.
- Designed and operated the **AI document-intelligence pipeline** — re-architected a monolithic worker into isolated, autoscaling stages (parsing, OCR, ML extraction, post-processing) handling **20M+ daily requests**, with **confidence scoring, human-in-the-loop fallback, and accuracy-drift dashboards** to catch silent failures before they reached enterprise customers.
- Successfully led **SOC 2 Type I and Type II audits**, maintaining a compliance score **above 95% year-round**. Set up MDM via **JumpCloud**, endpoint protection with **Bitdefender**, and launched a **Vulnerability Disclosure Program (VDP)** through **HackerOne**. Handled **customer security assessments** — fielding and responding to enterprise security questionnaires on engineering practices, architecture, and data handling.
- Owned product engineering, security & compliance, QA, and operational excellence; partnered with executive leadership to align technology strategy with business OKRs.
- Led **AWS cost optimization**, reducing monthly spend from **$288K to $70K/month** — a **76% reduction** through Savings Plans, Reserved Instances, and architecture redesign across ECS clusters and load balancers.
- Executed a comprehensive **AWS resource tagging strategy** to achieve per-feature cost attribution — enabling **COGS analysis**, per-product cost breakdowns, and environment-level cost visibility (dev, staging, production).
- Built production observability with **CloudWatch**, security posture with **AWS Inspector** and **GuardDuty**, and **ECS Auto Scaling** to ensure uptime SLAs defined in MSAs. Owned **incident response** — led triage, root cause analysis, and resolution across production systems.
- Architected and deployed an **end-to-end AI-powered agentic development pipeline** using Claude, increasing developer velocity by **80%+** — multi-agent orchestration with sub-agents and **human-in-the-loop quality gates** across code generation, review, testing, and deployment; integrated with **Jira, Confluence** (requirement ingestion) and **Figma** (design-to-code) via MCP; automated code review fixes, test generation with self-healing loops, and staging deployment with smoke tests and rollback.
- Built a strong **QA culture and testing discipline** — established unit, integration, and E2E testing practices; implemented test automation frameworks; defined quality gates for releases across the CI/CD pipeline.
- Conducted **200+ technical interviews in a single quarter** — highest in the organization — to rapidly scale the engineering team while maintaining a high hiring bar.
- Fostered a culture of ownership and accountability; mentored engineers and developed future engineering leaders.

Lead Engineer · Parspec Apr 2022 – Jan 2025

- Managed a team of **18 engineers**, including **Staff engineers** driving execution across frontend and backend while building the initial product foundation and establishing scalable architecture patterns.
- Owned high-impact projects end-to-end — concept to production — ensuring performance, reliability, and maintainability.
- Built and optimized systems for high-throughput, CPU-intensive workloads: document processing, generation, and data pipeline orchestration.

- Reduced **Alternate Product search engine** P90 latency for the top 500 manufacturers from **10 seconds to under 2 seconds**, significantly improving user experience and conversion.
- Reduced **Document search** P90 latency from **8 seconds to 2 seconds**, improving responsiveness for core product workflows.
- Architected and built a high-throughput **Inventory Management System** capable of ingesting and processing **10M+ products in a single bulk upload**, enabling enterprise-scale catalog onboarding with zero downtime.
- Introduced engineering best practices across code quality, testing, and deployment; mentored early team members and contributed to hiring.

Founding Engineer · Spotnana — Jul 2021 – Apr 2022

Bengaluru, India · Cloud-native B2B travel management platform

- Designed foundational backend systems powering end-to-end travel booking, management, and orchestration across **all travel verticals — Flight, Hotel, Rail, Car, and Limo**.
- Built deep integrations with **Sabre GDS** for reservations, ticketing, and global travel workflows across multiple verticals.
- Engineered robust, scalable APIs for complex travel data — itineraries, pricing, availability, and booking state transitions with strong consistency across distributed services.

Senior Software Engineer · Locus.sh — Jul 2020 – Jul 2021

Bengaluru, India · Enterprise logistics optimization platform

- Engineered distributed logistics systems serving enterprise customers — **Myntra, BigBasket, Licious, Delhivery, Bukalapak, Unilever** — processing **1.5–2M deliveries/day at BAU**, scaling to **5–8M deliveries/day** during peak sale events (Myntra EORS, Indonesia Harbolnas), generating **1B+ API requests/day**.
- Built and optimized compute-intensive **route optimization services** solving NP-hard problems with **180+ variables per decision** — dispatch planning, dynamic scheduling, and real-time ETA recalculations — achieving **10–20x improvement in trip efficiency** per vehicle.
- Designed high-throughput backend systems handling **peak bursts of 15,000–30,000+ RPS** across order ingestion, geocoding, real-time tracking, and driver dispatch — with route-optimization-heavy calls at **1,000–2,000 RPS** driving the bulk of compute cost.
- Built surge-resilient architecture using **queue-based decoupling, geospatial indexing, pre-computed route caching**, and **solver parallelization** to handle 3–4x traffic spikes during festive-season sales with zero downtime.

EARLIER EXPERIENCE

Senior Developer · CoWrks — Jan 2019 – Jul 2020

Bengaluru, India · Premium coworking platform (acquired by Brookfield in $2B deal) · 44 centers across 8 cities

- Built **Nucleus** (admin portal) and **Launchpad** (client platform) powering end-to-end coworking operations across **44 centers in 8 cities** — member onboarding, KYC, digital contracts with e-sign, payments and billing, space booking, and community management.
- Designed multi-tenant architecture supporting both individual members and enterprise clients with role-based access, automated invoicing, and real-time occupancy tracking across all centers.
- Implemented production observability with **Graylog** and **Errbit**, enabling proactive issue detection and reducing mean time to resolution across the platform.

Founding Engineer · Tekion Corp — Feb 2018 – Jan 2019

Bengaluru, India · Cloud-native Dealer Management System disrupting CDK/Reynolds duopoly · Founded by former Tesla CIO

- Joined at **seed stage** as a founding engineer — Tekion has since grown to a **$4B valuation**, serving **800+ dealerships** across **52 OEM brands** (GM, Ford, Honda).

- Designed and built core product modules — **Scheduling, Vehicle Inspection, and Promotions** — for the industry's first cloud-native DMS, replacing 40-year-old legacy on-premise systems.

Software Engineer · **HurixDigital** Feb 2016 – Feb 2018

Mumbai, India · AI-enabled digital content & publishing platform · Serving 15M+ users across 32 countries

- Developed interactive content delivery modules for global publishers including **McGraw Hill, Oxford University Press, and Pearson** — platform serves **15M+ users** across **32 countries** in **25+ languages**.

TECHNICAL SKILLS

Languages	Python, JavaScript (ES6+), TypeScript, Go
Backend	Django, Node.js, Express.js, GraphQL, Distributed Systems, Microservices
AI / Agents	Agent Orchestration, Multi-Agent Systems, Autonomous Agents, LangChain, LangGraph, LangFuse, CrewAI, LLM Tool/Function Calling, Structured Outputs, RAG, Agent Evals & Observability, Intent/Reply Classification, Human-in-the-Loop Guardrails, Persistent & Shared Agent Memory, Prompt Engineering, Voice Agents, MCP (Jira, Confluence, Figma), Claude / OpenRouter, OCR & ML Extraction, Confidence Scoring, Agentic CI/CD
Frontend	React, Redux, Next.js
Databases	PostgreSQL, MongoDB, Redis, Elasticsearch, DynamoDB
Messaging	AWS SQS, AWS SNS, AWS EventBridge, RabbitMQ, Kafka
Cloud & DevOps	AWS (ECS, CloudWatch, Inspector, GuardDuty, Auto Scaling, Load Balancers, Savings Plans, Cost Explorer), CI/CD Pipelines, Docker, Fluent Bit, Cost Optimization, Observability
Security	SOC 2 Type I & II, MDM (JumpCloud), Endpoint Protection (Bitdefender), VDP (HackerOne)
Leadership	Team Building (0→40+), OKR Management, Agile/Scrum, Technical Roadmapping, Hiring

EDUCATION

B.Tech — Gauhati University, Guwahati 2011 – 2015